

| Industrial Safety

- Industrial hazards and the organisation of safe work.
- Lifting equipment, pressure equipment and electrical installations.
- Accident prevention, worker actions and production control.

**A training course for
work in the Republic
of Kazakhstan.**

| What the course covers

- General requirements apply across production operations.
- Each equipment section covers documentation, authorisation and protective measures.
- Practical scenarios help identify conditions that require work to stop.

**Equipment-specific
authorisation is
arranged separately.**

| Purpose of industrial safety

- Industrial safety protects people, property and the environment from hazardous production factors.
- Accident prevention begins before equipment starts.
- Emergency preparedness includes communication, evacuation and rehearsed personnel actions.

Safety must be maintained throughout the equipment life cycle.

| Regulatory framework

- The Kazakhstan Law on Civil Protection establishes general requirements.
- Order No. 359 covers lifting equipment; Order No. 358 covers pressure equipment.
- Electrical safety and hot work are governed by their specific rules.
- Work instructions account for the equipment and site conditions.

A sample instruction cannot replace requirements for the actual task.



Hazardous production facilities

- Hazardous substances and processes require checking the criteria for a hazardous facility.
- Identification follows legislation rather than the everyday name of a business.
- Hazardous technical devices are assessed according to their type and parameters.
- Workers must know their area's hazards and the limits of their authorisation.

A facility name without the word hazardous does not remove the risk.

Hazardous substances and processes

- Combustible substances can ignite; oxidisers intensify combustion.
- Toxic substances are hazardous through inhalation and contact.
- Pressure, temperature and mechanical energy can cause severe injury.
- Work requires information about the substance, process and protective measures.

Hazards depend on properties and handling conditions.

| Accidents and incidents

- Destruction, an uncontrolled explosion or a hazardous release can indicate an accident.
- Equipment failure or a process deviation can constitute an incident.
- An event must be reported and examined even if nobody is injured.
- Concealing a deviation prevents the organisation from averting a later accident.

**Record deviations
before they recur.**

Duties of the organisation

- The organisation maintains equipment and meets mandatory requirements.
- It appoints responsible persons and establishes control procedures.
- It arranges training, knowledge assessment and personnel authorisation.
- It maintains readiness to contain accidents and deal with their consequences.

Appointing a responsible person does not remove the organisation's duties.

| Worker actions

- Workers perform assigned tasks within their training and authorisation.
- They check the workplace and use the specified protective equipment.
- They report defects, deviations and threats to people.
- In immediate danger, they stop work and move to safety.

**Do not attempt an
unauthorised repair
of a dangerous fault.**

| Training and authorisation

- Training explains hazards and safe work methods.
- Knowledge assessment verifies understanding of requirements.
- A briefing addresses the actual task and workplace.
- Practical preparation and authorisation are required where stipulated by the rules and task.

Viewing a presentation does not authorise every operation.

Production control

- Control identifies breaches before an accident occurs.
- Checks cover equipment, documentation, personnel and working conditions.
- Each finding receives an action, an owner and a deadline.
- Closure requires verification that the issue has actually been corrected.

**A closed finding
needs confirmation
on site.**

Checks before starting work

- Clarify the task, work-zone boundaries and responsible persons.
- Check equipment, protective devices and clear access routes.
- Assess neighbouring activities and possible changes in conditions.
- Do not begin until unsuitable conditions have been corrected.

Yesterday's good condition does not replace today's check.

Workplace risk assessment

- Identify what could cause harm and who could enter the danger zone.
- Prefer hazard elimination and engineering protection.
- Organisational measures and PPE provide additional protection.
- Repeat the assessment when equipment, tasks or surroundings change.

**Changed conditions
require a new
assessment.**

Work permits and coordination

- Use a work permit where required by the rules and established procedures.
- Specify the location, task, personnel and protective measures.
- Actual workplace conditions must match the conditions of authorisation.
- A changed task requires review of the permit conditions, not an informal agreement.

A signature on a permit does not replace completed protective measures.

| Isolation of energy sources

- Before maintenance, identify electrical, mechanical and stored energy.
- Stop equipment and prevent unintended starting.
- Release pressure and secure moving parts under the established procedure.
- Authorised personnel verify the absence of hazardous energy.

**The Stop button
alone does not
isolate energy.**

| Hot work

- Welding, cutting and heating produce sparks and hot surfaces.
- Remove combustible materials or protect them using the specified method.
- Check adjacent rooms, openings and concealed spaces.
- Provide firefighting equipment and post-work monitoring.

**Sparks can travel
beyond the visible
work area.**

| Equipment documentation

- The equipment passport identifies the device and its permitted parameters.
- The manufacturer's manual defines operating and maintenance conditions.
- Logs preserve inspection, repair and defect records.
- Establish unknown parameters before operating equipment.

A nearby device's parameters cannot replace the equipment passport.

Lifting equipment

- Cranes, hoists and other lifting devices have different applications.
- Slings, lifting beams and containers form part of the lifting system.
- Equipment selection depends on load mass, the work zone and installation conditions.
- Plan the operation using the equipment passport and lifting procedure.

Safe lifting depends on the entire system.

Lifting equipment passport and markings

- Check the equipment type and identification number.
- Verify lifting capacity, installation conditions and manufacturer restrictions.
- Check inspection records and authorisation for use.
- Clarify illegible or contradictory markings.

Verify lifting capacity for the actual operating conditions.

| Recording and maintaining lifting equipment

- State registration or internal recording requirements depend on the device type.
- Exemption from state registration does not remove control obligations.
- Maintain and repair equipment according to the established schedule.
- Document inspections and the correction of defects.

Internal records also require clear responsibility.

| Responsibilities during crane operations

- Supervision oversees safe operation.
- The person responsible for serviceable condition organises maintenance and repair.
- The person responsible for load handling organises the safe lifting operation.
- The crane operator and slinger fulfil their duties and use agreed signals.

**Responsibilities
must be understood
before lifting begins.**

| Preparing a lift

- Determine the load mass, centre of gravity and attachment points.
- Select equipment and accessories with suitable characteristics.
- Check the route, landing area and absence of people.
- Agree on communication between participants.

**Establish an
unknown load mass
before lifting.**

| Slings and lifting accessories

- Check identification and suitability for the chosen lifting arrangement.
- Inspect ropes, chains, sling legs, hooks and connections.
- Remove damaged or unidentifiable accessories from service.
- Protect load edges using the specified padding.

A knot in a sling is not a repair.

Lifting danger zone

- Keep people out of the area where a load could fall or move.
- Do not allow anyone beneath a suspended load.
- Do not guide a load by hand while positioned between it and an obstacle.
- Maintain the zone boundaries throughout the operation.

A short lift presents the same hazard.



Signals and stopping

- Agree on clear signals and the communication method in advance.
- Stop movement if communication is lost.
- A stop signal concerning danger to people requires an immediate response.
- Resume only after the cause is resolved and actions are coordinated.

Do not guess the meaning of an unclear signal.

| Inspection of lifting equipment

- Shift inspections, maintenance and technical examinations serve different purposes.
- Check the structure, mechanisms, brakes and protective devices.
- Determine intervals and scope from the applicable rules and equipment documentation.
- Do not bypass a defective limiter to finish a shift.

**Protective devices
are part of the
conditions for safe
operation.**

| Testing and repairing lifting equipment

- Qualified specialists conduct tests under the established programme.
- A test load does not authorise overloads during normal operation.
- Certain repairs or modifications require additional examination.
- Record the results and authorisation for continued operation.

**A trial lift cannot
replace the required
post-repair
examination.**

| Welded joints in lifting equipment

- Repair safety-critical components using an established procedure.
- Determine welder, material and inspection requirements for the structure.
- Cracks, deformation and unacceptable defects require specialist assessment.
- Unauthorised welding can alter structural behaviour.

Repair of a load-bearing structure requires a specific decision.

| Scenario: a damaged sling

- A sling is defective, but the load must be moved urgently.
- Do not begin the lift; notify the responsible person.
- Prevent further issue of the accessory pending the required decision.
- Continue with suitable, serviceable equipment.

**Urgency does not
change acceptance
criteria.**

| Pressure equipment

- Boilers, vessels, pipelines and cylinders contain stored energy.
- Loss of containment can cause a release, burns or harmful exposure.
- Risk depends on pressure, temperature and the working medium.
- Operate each device within its established limits.

A small vessel is not necessarily a low-hazard vessel.

| Operating and design parameters

- Operating, design, permitted and test pressures serve different purposes.
- Test pressure applies to an established test procedure.
- Use the documented and authorised operating limits.
- Observe temperature restrictions and medium compatibility.

A test parameter is not an operating parameter.

Vessel classification and applicable rules

- Classification depends on parameters and the nature of the medium.
- Pressure alone is insufficient for a universal assessment.
- A specialist determines applicable requirements and inspection scope.
- Operators use the established restrictions for their own equipment.

**Classification
affects inspection
and documentation.**

| Protective devices

- Safety devices limit dangerous parameter exceedances.
- Their condition and settings require established checks.
- Do not bypass, blank off or alter protection without authorisation.
- Investigate the cause of a protective trip.

Protection is not intended to compensate continually for an incorrect operating mode.

| Monitoring pressure and temperature

- Compare instrument readings with the permitted operating conditions.
- Check for visible damage and unreliable readings.
- Replace a faulty instrument under the established procedure.
- Respond to leakage, vibration or unusual noise as instructed.

One normal reading is insufficient when other danger signs are present.

| Starting and monitoring operation

- Confirm equipment readiness and authorisation before starting.
- Check valves, protection and the required instruments.
- Change parameters in the sequence specified by the instructions.
- Respond to deviations as instructed; do not guess settings.

**Starting after repair
requires confirmed
readiness.**

| Stopping and preparing for maintenance

- Isolate equipment from pressure sources and hazardous media.
- Release pressure and verify a safe condition.
- Clean, purge and test the atmosphere where required.
- Begin work after completing protective measures and authorisation.

**A faulty gauge
reading zero does
not prove safety.**

Technical examination of vessels

- Inspection and testing assess technical condition.
- Intervals depend on the device and applicable requirements.
- Do not start work when an examination is overdue; notify the responsible person to arrange the examination and authorisation.
- Record the results in the equipment documentation.

Do not copy an inspection interval from a different vessel.

| Compressed gas cylinders

- Identify the gas using markings and documentation.
- Protect cylinders against falling, heat and valve damage.
- Use compatible regulators and serviceable connections.
- Never check leaks with an open flame.

Do not connect a cylinder with uncertain identification.

| Repairing and welding vessels

- Repair only after safe removal from service.
- The procedure, personnel and inspection must suit the vessel's purpose.
- Do not weld a container with unidentified residual contents.
- Return to service requires documented inspection results.

An apparently empty container may retain explosive residues.

| Scenario: rising pressure

- Pressure rises and normal control fails to restore the operating conditions.
- The operator follows the emergency shutdown instructions.
- Move people out of danger and notify the responsible person.
- Do not bypass protection or repair equipment under pressure.

Investigate the cause after making the situation safe.

| Electrical hazards

- Current, electrical arcs and hot components can cause severe harm.
- Moisture, damaged insulation and conductive surfaces increase danger.
- The absence of sparks or noise does not establish absence of voltage.
- Restrict access to electrical installations under the established procedure.

Voltage can remain dangerous without any advance sensation.

| Personnel categories and authority

- Duties and electrical safety qualification groups determine permitted actions.
- A group does not authorise every task without an assignment and authorisation.
- Only authorised personnel connect, repair and switch equipment.
- Learners must remain within the limits of their preparation.

**Knowing a term
does not authorise
an operation.**

| Organising work in electrical installations

- Determine the required form of authorisation for the specific task.
- Appoint responsible persons and the work team.
- The person admitting the team checks workplace preparation and gives a briefing.
- Record breaks, relocation and completion under the established procedure.

**The actual work
must match the
assigned task.**

| Work with voltage removed

- Authorised personnel carry out the necessary disconnections.
- They prevent unintended energisation and display prohibition notices.
- They verify absence of voltage and install the required earthing.
- They install the necessary barriers and other notices.

Disconnecting and verifying absence of voltage are different actions.

| Earthing and protective conductors

- The protective conductor is part of the shock-protection system.
- Repair damaged connections before work.
- Do not replace a protective conductor with an improvised metal connection.
- Specialists carry out checks and repairs.

An intact-looking enclosure does not prove effective protection.

| Portable equipment

- Inspect the enclosure, cable, plug and connections before use.
- Select equipment for the environment and follow its instructions.
- Stop using equipment with a damaged cable or signs of overheating.
- An improvised connection does not replace a serviceable assembly.

**Check condition
before switching on.**

| Electrical protective equipment

- Select protection for the voltage and nature of the work.
- Before use, check integrity and required test markings.
- Do not use damaged equipment or equipment with overdue tests.
- PPE does not permit bypassing organisational measures.

An unknown test date requires verification.

| Work near live parts

- Determine boundaries and safe distances for the actual installation.
- Maintain them for tools, loads and equipment parts as well as people.
- Do not move barriers without authorisation.
- If conditions change, stop work until protective measures are reviewed.

**Dangerous
approach does not
require direct
contact.**

| Returning electrical equipment to service

- Verify completion and removal of people from the danger zone.
- Remove temporary protective measures in the established sequence.
- Record completion and authorisation to energise.
- Authorised personnel carry out energisation.

A trial start requires the same safety controls.

| Scenario: a damaged cable

- A portable tool has an exposed conductor.
- Stop using it and prevent further energisation.
- Notify the responsible person to arrange repair or replacement.
- Do not touch the damage or repair the cable while energised.

**Applying tape alone
does not authorise
continued use.**

| Emergency notification

- Report the location, hazard and known information about casualties.
- Use the site's established communication channels; call 112 or 101 where needed.
- Do not delay reporting while independently searching for the cause.
- Warn nearby people without exposing yourself to danger.

The report must enable responders to locate the danger quickly.

| Evacuation and assembly point

- Use a safe escape route appropriate to the hazard.
- Do not return for belongings or enter a closed-off zone.
- At the assembly point, report your presence and known missing persons.
- Return only when authorised after conditions have been assessed.

**Report immediately
if a person's
whereabouts are
unknown.**

| Safety when giving assistance

- First make sure the rescuer is not exposed to danger.
- Call emergency services and describe the situation.
- Act within your training and follow the dispatcher's instructions.
- Do not enter a hazardous atmosphere or touch an energised casualty without appropriate protection.

**A second casualty
makes rescue more
difficult.**

| Reviewing an event and restarting

- Preserve information about signs, timing and circumstances.
- Establish causes from facts, including organisational conditions.
- Verify corrective measures before resuming work.
- Inform personnel about changes to instructions and equipment.

Restart requires correction of the cause, not just the consequences.

| Final knowledge assessment

- Work requires a clear task, serviceable equipment and valid authorisation.
- Lifting defects, pressure leaks and electrical damage require timely stopping.
- Emergency actions begin with protecting people and raising the alarm.
- Select Start test.

If conditions are uncertain, wait for verification before working.